

User Acceptance Testing (UAT)

SOFTWARE VALIDATION & QUALITY ASSURANCE REPORT

Document Ref: UAT-SWTID-2026-8956

Date	29/09/26	Team ID	SWTID-2026-8956
Project Name	WeatherWise AI	Maximum Marks	[Enter Marks]

Project Team Details (Team Size: 4)

Team Leader: SANMATHI KUMAR

Team Member: ARIFA ZOHRA

Team Member: AINUL LIYANA P M

Team Member: VAISHALI S

1. PROJECT OVERVIEW

System Name	WeatherWise AI Backend Service	Version	v1.0.0
Testing Period	25/09/26 to 29/09/26	Architecture	Node.js / Express / MongoDB (MVC)

Description: WeatherWise AI is a RESTful backend service designed to deliver real-time weather metrics paired with AI-generated natural language summaries and activity/clothing recommendations. Built on the Node.js/Express framework, it incorporates Google Gemini AI (@google/genai) and OpenWeatherMap APIs to deliver tailored advice based on live environmental context.

2. TESTING SCOPE

- User Auth (Register, Login, bcryptjs hashing, JWT)
- Protected Endpoint Access & JWT Middleware
- Location Management & Unique Compound Indexes
- OpenWeatherMap API Metrics Fetching
- Google Gemini AI Recommendation Engine
- Fallback Execution & NoSQL Sanitization

3. TESTING ENVIRONMENT

- URL / Location:** http://localhost:5000/api
- Tools:** Postman / Thunder Client
- Runtime:** Node.js v16+, Express v4
- Database:** MongoDB via Mongoose ODM
- Test Credentials:**
 - Admin: admin@weatherwise.ai
 - User: user@weatherwise.ai

4. TEST CASES EXECUTION LOG

TC ID	Scenario	Test Steps	Expected Result	Status
TC-001	User Registration	1. POST /api/auth/register with valid payload. 2. Inspect MongoDB collection.	Account created; password correctly hashed via bcryptjs.	PASS
TC-002	Authentication & JWT	1. POST /api/auth/login with credentials. 2. Validate response header/body.	Returns HTTP 200 with signed bearer JWT token.	PASS
TC-003	Save Favorite Location	1. Attach JWT header. 2. POST /api/locations with city/country.	Location saved in DB linked to user ID foreign key.	PASS
TC-004	Compound Index Unique Check	1. Re-send POST /api/locations with same city/country for user.	Database rejects duplicate entry via compound index.	PASS

4. TEST CASES EXECUTION LOG (CONTINUED)

TC ID	Scenario	Test Steps	Expected Result	Status
TC-005	Public Weather Metrics	1. GET /api/weather/:city without auth header.	Returns live weather metrics from OpenWeatherMap API.	PASS
TC-006	Gemini AI Insights	1. Send auth JWT. 2. POST /api/ai/weather-recommendation.	Structured JSON with clothing & activity advice.	PASS
TC-007	Fallback Mode Security	1. Revoke API keys. 2. Request weather and AI endpoints.	Graceful execution using internal fallbacks without system crash.	PASS

5. BUG TRACKING & RESOLUTION

Bug ID	Description	Steps to Reproduce	Severity	Status	Resolution
BG-001	Unsanitized input permitted potential NoSQL parameter injection on login.	1. POST /api/auth/login with MongoDB query operators (e.g., \$gt).	High	Closed	Enforced global sanitization middleware across Express routes.
BG-002	Gemini API network timeout under rapid batch calls caused unhandled rejection.	1. Fire 20 concurrent AI recommendation requests simultaneously.	Medium	Closed	Wrapped external service calls with timeout handling and try-catch fallbacks.

6. VERIFICATION & SIGN-OFF

Lead Tester: SANMATHI KUMAR Role: Team Leader Date: 29/09/26	Project Manager: [Manager Name] Role: Project Manager Date: 29/09/26	Product Owner: [Owner Name] Role: Product Owner Date: 29/09/26
<u>Sanmathi Kumar</u>	<u>[Pending Verification]</u>	<u>[Pending Deployment]</u>

UAT Execution Notes & Principles:

- All functional pathways (positive auth, validation errors, duplicate checks) were thoroughly verified against the MVC system specifications.
- Security auditing confirmed protection against XSS and NoSQL injection attacks on public and guarded routes.
- Formal approval from both Project Manager and Product Owner is required prior to production deployment.

